Drew Guardiola

drewguardiola@u.northwestern.edu | drewguardiola.tech | (201) - 575 - 6534 | LinkedIn | Glen Ridge, NJ

EDUCATION

Northwestern University - McCormick School of Engineering **Evanston, IL**
Combined B.S./M.S. in Mechanical Engineering - Aerospace Specialization, GPA: 3.75/4.00 *Expected December 2027*
Relevant Coursework: Engineering Analysis I-IV, Design Thinking and Communication, Electronics Design, Mechanics of Materials, Mechatronics, Thermodynamics, Mechanical Design, Fluid Mechanics, Machine Dynamics, Embedded Programming in Python

TECHNICAL EXPERIENCE

Honeywell Aerospace Technologies **South Bend, IN/Phoenix, AZ**
Mechanical Engineering Intern - South Bend, IN *June 2026 - September 2026 (current)*

- Authoring 30+ Customer Engineering documentation packages (Conformance, Engineering Reports, Usage Guides, Drawing Markups) supporting the Poland AGT1500 depot project for the U.S. Army's M1 Abrams turbine engine
- Developing 13 SOWs defining scope, responsibilities, and acceptance criteria for Airbus A350 test stand procurement
- Supporting Factory Acceptance Testing of the F124 turbofan Main Fuel Control electrical tester; providing engineering disposition on quality feedback including damage analysis from technical drawings and CSE manual support

Systems Engineering Intern - Phoenix, AZ *June 2025 - September 2025*

- Set up and ran FEA and CFD simulations for propulsion systems using physics-based analysis and design validation
- Verified 200+ engine requirements and enabled team-friendly reporting through 50+ reports in DOORS Report Builder

PROJECTS

Tesla Valve CFD Analysis **Evanston, IL**
Self-Directed CFD Project (ME378) *Summer 2026*

- Simulated forward and reverse flow through a twelve-stage 2D Tesla valve in ANSYS Fluent across a Reynolds number sweep, capturing the inertial onset of the diode effect near Re=200 as predicted by Nguyen et al. (2021)
- Measured diodicity rising from ~1 below onset to 1.78 at Re=1000 by combining a laminar steady solver below onset with laminar transient and time-averaged ΔP above onset, verified via a three-mesh GCI study across multiple geometries

Northwestern Formula Racing **Evanston, IL**
Aerodynamics Subteam Engineer *September 2024 - Present*

- Optimized vehicle performance through CFD, FEA, and iterative CAD design, increasing the lift/drag ratio from 1.89 to 2.03
- Optimized geometry of the rear wing endplates and airfoils, ensuring compliance with FSAE competition regulations

3D-Printed Bicycle Brake Caliper **Evanston, IL**
Mechanical Design (ME240) *Spring Quarter 2025*

- Leveraged Siemens NX to utilize topology optimization and FEA for weight and strength analysis to model a brake caliper
- Designed a 3D-printed bicycle brake caliper using load pathing, stress analysis, failure theories, and engineering first principles to meet strength and weight goals, allowing a bicycle traveling at 20 miles per hour to stop within 5 meters

NASA Dragonfly Rotorcraft Hover Analysis **Evanston, IL**
Computational Aerospace Analysis (ME364) *Spring Quarter 2026*

- Modeled NASA Dragonfly rotorcraft hover aerodynamics in Titan's atmosphere using ANSYS Fluent CFD (k-ω SST turbulence, C-mesh, compressibility) coupled with momentum theory across angles of attack to find peak C_l/C_d of 27.22
- Computed 4.3kW total hover power, validating Dragonfly's charge-and-hop mission architecture given MMRTG output

Automatic Card Shuffler **Evanston, IL**
Self-directed Capstone Project *Spring Quarter 2024*

- Designed and built an autonomous card shuffler by integrating DC motors, microcontroller logic, and mechanical parts such as laser-cut acrylic and CNC routed-wood component for reliable and effective hands-free operation
- Required card deck to be processed only twice before it was thoroughly shuffled, reducing shuffling time by around 20%
- Managed timeline and set goals autonomously, led mentor check-ins, presented functional prototype to peers

AWARDS + ACTIVITIES

Susquehanna International Group Poker Tournament 2024-2025 **Chicago, IL + Philadelphia, PA**

- Placed 2nd out of 100, winning $1,000 by applying advanced critical thinking skills competing against 500+ qualifiers
- Placed 5th out of 8,000, winning $1,500 after qualifying through regionals and outperforming in SIG's national tournament

SKILLS AND INTERESTS

Engineering/Computer: Ansys, Solidworks/Siemens NX - 3D CAD, 3D Printing - FDM, CNC Router, GD&T, DFM/DFA, Python
Systems: DOORS, PLM/ECN Workflows, Technical Writing, Statements of Work, Microsoft Excel, MATLAB
Interests: Texas Hold 'em, Tenor Saxophone, Tennis, Biking, Weightlifting, Snowboarding, Photography, Puzzles